

Database Management System

BCA — Semester IV — 2023 — Answer Sheet

Subject Code: CACS255

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Define database and DBMS. Explain the benefits of DBMS in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are different SQL constraints to specify the rules for the data in the database? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is relational algebra? Explain different RA operations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. How candidate key is different from primary key? Also explain foreign key in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is a stored procedure? How can it be used in a database? Explain with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What are the major purposes of concurrency control? Explain two phase locking protocol in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is database security? Describe steps of query processing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What are fully and partial functional dependencies? Explain 2NF and 3NF in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How different join operations can be performed? Explain with proper examples. Consider a database of a shipment company. Shipped items are received into the system at a single retail center. Shipped items make their way to their destination via one or more standard transportation events (i.e. flights, truck deliveries). The transportation events follow a regular schedule. Create an ER diagram to represent this information.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Consider the following tables: Book(Bookid, Title, Publisher, Published Date, Price)Book_Authors (Bookid, Author name)Borrower (Card No, Name, Address, Contact)Bookissue (BookId, CardNo, IssuedDate, Due Date)Write SQL query for the following:a) Find the name of the books taken by Ramesh Sharma.b) Find the Issued book name and the borrower's name who have Issued date before 15th May, 2023 and due date after 15th June, 2023.c) Find the borrower's name who has taken the book written by Laxmi Prashad Devkota.d) Find the number of books Issued on the date 25th April, 2023.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.